



ZXA10 C300

Optical Access Convergence Equipment

Hardware Description

Version: V2.0.0

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About This Manual

Purpose

The ZXA10 C300 Optical Access Convergence Equipment (ZXA10 C300 for short) is a full-service optical access platform. It supports the video, data, voice, TDM, and CATV services. The ZXA10 C300 can be connected with MDUs/ONUs/ONTs through various networking technologies.

This manual provides hardware introduction that support the installation of the ZXA10 C300 Optical Access Convergence Equipment.

Intended Audience

This document is intended for:

- Planning engineer
- Installation supervision engineer
- Debugging engineer

What Is in This Manual

This manual contains the following chapters:

Chapter	Summary
1, Shelves	Describes the shelves structure and configuration.
2, Cards	Describes the cards supported by the ZXA10 C300.

Conventions

This manual uses the following typographical conventions:

Typeface	Meaning
	Note: provides additional information about a certain topic.

Chapter 1

Shelves

The ZXA10 C300 uses the following two types of shelves:

- ETSI 21-inch shelf
- IEC 19-inch shelf

A 19-inch shelf and a 21-inch shelf share the same cards excluding backplanes and fans.

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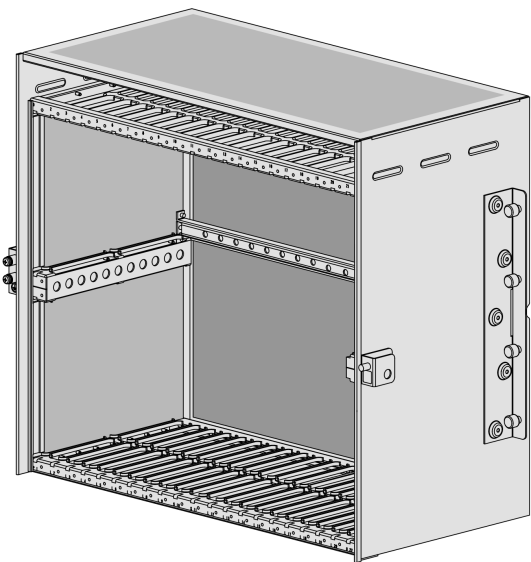
ETSI 21-Inch Shelf	1-1
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1.1 ETSI 21-Inch Shelf

Structure

The 21-inch shelf is installed in a B6030-22C-EB cabinet. [Figure 1-1](#) shows the 21-inch shelf structure.

Figure 1-1 21-Inch Shelf Structure



Configuration

The shelf is divided into two areas: service card area (lower) and fan area (upper, optional).

- Service card area

The 9U-height area is used for switching and control cards and service cards installation.

Width of the slots 0/1/10/11/21/22 is 25 mm, while width of other slots is 22.5 mm.

Figure 1-2 shows the 21-inch shelf configuration.

Figure 1-2 21-Inch Shelf Configuration

0	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Power Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Switching and Control Card	Switching and Control Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Common Interface Card	Uplink Card
1																				
Power Card																				

Table 1-1 lists the shelf configuration description.

Table 1-1 Shelf Configuration Description

Slot Number	Card Type
0/1	Front-outlet power card
2 to 9	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
10/11	Switching and control card
12 to 19	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
20	Common interface card
21/22	Uplink card

- Fan area

The fan area is a plug-in card with 1U height and 21-inch width.

It provides the system with cool wind in ventilating way. Fan speed can be adjusted according to the system temperature.

Technical Specifications

Table 1-2 lists technical specifications of the 21-inch shelf.

Table 1-2 21-Inch Shelf Technical Specifications

Item	Specification
Dimensions	449.2 mm × 535 mm × 270 mm (Height × Width × Depth)

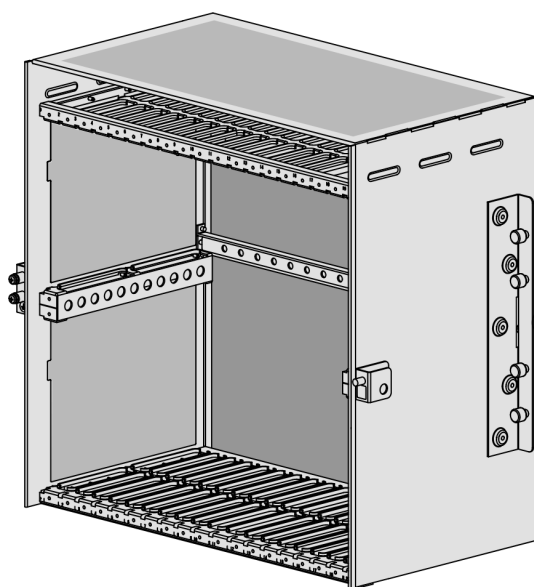
Item	Specification
Weight	14 kg (Empty) 36.6 kg (Full configuration)

1.2 IEC 19-Inch Shelf

Structure

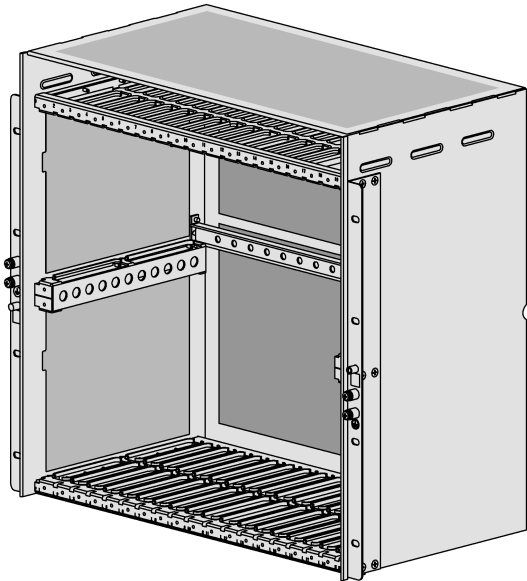
When the 19-inch shelf is installed in a B6030-22C-IB cabinet, its mounting ears are close to the back edge, see [Figure 1-3](#).

Figure 1-3 19-inch Shelf Structure (Back Mounting Ears)



When the 19-inch shelf is installed in a B6060-22F cabinet, its mounting ears are close to the front edge, see [Figure 1-4](#).

Figure 1-4 19-Inch Shelf Structure (Front Mounting Ears)



Configuration

The shelf is divided into two areas: service card area (lower) and fan area (upper, optional).

- Service card area
The 9U-height area is used for switching and control cards and service cards installation.
Width of the slots 0/1/10/11/19/20 is 25 mm, while width of other slots is 22.5 mm.

Figure 1-5 shows the 19-inch shelf configuration.

Figure 1-5 19-Inch Shelf Configuration

0	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Power Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Switching and Control Card	Switching and Control Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Common Interface Card	Uplink Card
Power Card																		20
																		Uplink Card

Table 1-3 lists the shelf configuration description.

Table 1-3 Shelf Configuration Description

Slot Number	Card Type
0/1	Front-outlet power card

Slot Number	Card Type
2 to 9	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
10/11	Switching and control card
12 to 17	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
18	Common interface card
19/20	Uplink card

- Fan area

The fan area is a plug-in card with 1U height and 19-inch width.

It provides the system with cool wind in ventilating way. Fan speed can be adjusted according to the system temperature.

Technical Specifications

Table 1-4 lists technical specifications of the 19-inch shelf.

Table 1-4 19-Inch Shelf Technical Specifications

Item	Specification
Dimensions	443.7 mm × 482.6 mm × 270 mm (Height × Width × Depth)
Weight	12.5 kg (Empty) 34.2 kg (Full configuration)

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Chapter 2

Cards

The ZXA10 C300 shelf is composed of cards and a fan box. [Table 2-1](#) lists the cards and fan box.

Table 2-1 Card List

Card Type	Name	Description	Interface
Switching and control card	SCTM	TM-level switching and control card	1 out-of-band NM interface 1 serial interface 1 SD card interface 4 10GE/GE uplink interfaces 1 mirroring interface
	SCXN	Type-N switching and control card	
PON interface card	GTGH	16-port GPON interface card	16 GPON interfaces
	GTGO	8-port GPON interface card	8 GPON interfaces
	GTXO	8-port XG-PON1 interface card	8 XG-PON1 interfaces
	GTXQ	4-port XG-PON1 interface card	4 XG-PON1 interfaces
Ethernet uplink card	XUTQ	4-port 10GE optical interface Ethernet uplink card	4 10GE optical interfaces
	XUVQ	4-port 10GE optical interface Ethernet uplink card, supports SynE and IEEE 1588.	4 10GE optical interfaces
	HUKQ	2-port 10GE and 2-port 40GE optical interface Ethernet uplink card, supports SynE	2 10GE and 2 40GE optical interfaces
	HUTQ	2-port 10GE and 2-port GE optical interface Ethernet uplink card	2 10GE and 2 GE optical interfaces
	HUVQ	2-port 10GE and 2-port GE optical interface Ethernet uplink card, supports SynE and IEEE 1588.	2 10GE and 2 GE optical interfaces
	HUGQ	2-port GE and 2-port FE optical interface Ethernet uplink card	2 GE and 2 FE optical interfaces
	GUFQ	4-port GE optical interface Ethernet uplink card	4 GE optical interfaces
	GUTQ	4-port GE electrical interface Ethernet uplink card	4 GE electrical interfaces

Card Type	Name	Description	Interface
	GUSQ	2-port GE optical and 2-port GE electrical interface Ethernet uplink card	2 GE optical and 2 GE electrical interfaces
Ethernet interface card	GDFO	8-port GE optical Ethernet interface card	8 GE/FE optical interfaces
	HDTT	8-port GE and 2-port 10GE optical Ethernet interface card	8 GE/FE and 2 10GE optical interfaces
P2P interface card	FTGK	48-port P2P interface card	48 GE/FE optical interfaces
TDM interface card	CTLA	STM-N CES interface card	2 STM-1 or 1 STM-4 interface
	CTUB	32-channel E1 unbalanced CES interface card	32 E1 unbalanced interfaces
Common interface card	CICK	Common interface card, supports clock/time processing, environment monitoring, and OAM.	● 2 BITS/1PPS+TOD clock input interface ● 1 BITS/1PPS+TOD clock output interface ● 1 out-of-band NM interface ● 1 pulic serial port ● 1 temperature sensor interface ● 1 humid sensor interface ● 1 smog sensor interface ● 1 liquid sensor interface ● 1 entrance control sensor interface ● 1 digit input interface ● 1 digit output interface
Backplane	MWIA/R	19-inch backplane	-
	MWIA		
	MWEA/R	21-inch backplane	-
	MWEA		
Power card	PRWG	4.5U power interface card	● 1 power interface ● 2 RJ-45 interfaces (reserved)

Card Type	Name	Description	Interface
Fan box	FAN-19 (FCWB)	Standard 19-inch fan box, contains 3 fans.	-
	FAN-19 (FCWB)	Advanced 19-inch fan box, contains 10 fans.	-
	FAN-21 (FCWD)	Standard 21-inch fan box, contains 4 fans.	-
	FAN-21 (FCWD)	Advanced 21-inch fan box, contains 10 fans.	-

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2.1 Switching and Control Cards

Overview

The switching and control cards are responsible for system control and management, implement the non-blocking switching for all service cards in the ZXA10 C300.

The ZXA10 C300 supports the following switching and control cards:

- SCTM: supports 1.28 Tbps switching capacity and 288 K MAC address table.
- SCXN: supports 480 Gbps switching capacity, 64 K MAC address table, and VoIP function (with VoIP daughter-card).

Panel

Figure 2-1 and Figure 2-2 show the switching and control card panels.

Figure 2-1 SCTM Panel



Figure 2-2 SCXN Panel



Indicators

Table 2-2 describes the indicators of the switching and control cards.

Table 2-2 Indicator Description for Switching and Control Cards

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off.
	Green LED flashes slowly (0.5 Hz).	The card works properly and card type configuration is correct.
	Green LED is ON.	The card is not configured properly.
	Green LED flashes quickly.	The card is downloading data.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED is ON.	The card is in booting process.
M/S	Green LED is ON.	This card is the active card.
	LED is OFF.	This card is the backup card.
HDD	Red LED is ON.	The hard disk is busy. Card plug-in/plugin-out is not allowed.
	LED is OFF.	The hard disk is idle.
ACT <i>i</i> (<i>i</i> = 1–4)	Green LED is OFF.	No. <i>i</i> GE optical interface is not connected.
	Green LED is ON.	No. <i>i</i> GE optical interface is connected (LINK).
	Green LED flashes.	No. <i>i</i> GE optical interface is sending and receiving the data (ACTIVE).

Interfaces

Table 2-3 describes the interfaces of the switching and control cards.

Table 2-3 Interface Description for Switching and Control Cards

Interface	Description
GE1–GE4/XG1–XG4	GE/10GE optical interface, for uplink
SD	SD socket, for importing/exporting data
CLI	RS-232 port, for local debugging
10/100M	RJ45 interface, for out-of-band network management
DIAG	RJ45 interface, for mirroring service traffic

Button

The **RST** button is used for resetting card.

Principle Diagram

Figure 2-3 shows the principle diagram of the switching and control cards.

Figure 2-3 Principle Diagram of Switching and Control Cards

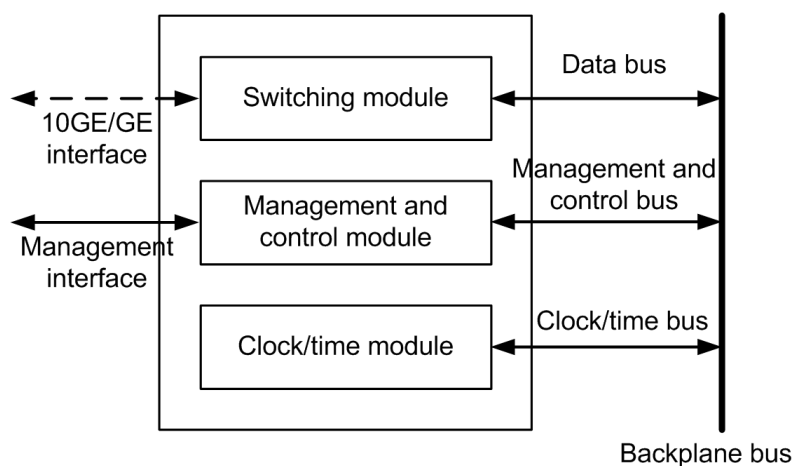


Table 2-4 describes the module functions of the switching and control cards.

Table 2-4 Module Functions of Switching and Control Cards

Module	Function
Switching module	The switching center of data service, implements data switching, Ethernet, VLAN, multicast, IP route, QoS, and security functions.
Management and control module	Consists of control software, protocol processing software, overhead processing, Ethernet switching chip, and control CPU, provides interfaces for cards to forcedly download versions and reset hardware/software , provides online detection for cards, implements fan detection and control functions.
Clock/time module	Processes system clock and time, complies with ITU-T G.8262, G.8264, G.781, and RFC 1305.

Technical Specifications

Table 2-5 lists technical specifications of the switching and control cards.

Table 2-5 Technical Specifications of Switching and Control Cards

Item	Specification
Power consumption	SCTM (without uplink interfaces): 111 W (static), 172 W (maximum), 142 W (typical) SCTM (with uplink interfaces): 115 W (static), 175 W (maximum), 145 W (typical) SCXN (without uplink interfaces): 70 W SCXN (with uplink interfaces): 75 W
Dimensions	395.5 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	SCTM: 1.075 kg SCXN: 1.132 kg

- Static power consumption: the system is powered on and all user ports are deactivated.
- Maximum power consumption: the system is powered on and all user ports are activated.
- Typical power consumption: the system is powered on and 50% user ports are activated.

2.2 PON Interface Cards

The ZXA10 C300 uses the following **PON** interface cards:

- GPON interface cards: GTGH and GTGO
- XG-PON1 interface cards: GTXO and GTXQ

2.2.1 GPON Interface Cards

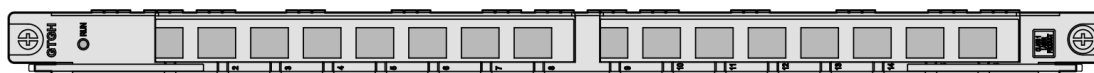
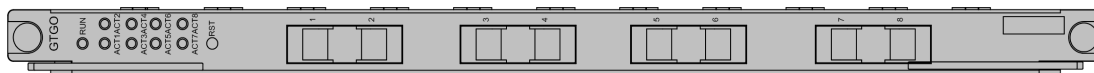
Overview

The GPON interface cards provide **GPON** access. The ZXA10 C300 supports the following GPON interface cards:

- GTGH: 16-port GPON interface card
- GTGO: 8-port GPON interface card

Panel

Figure 2-4 and Figure 2-5, and show the GPON interface card panels.

Figure 2-4 GTGH Panel**Figure 2-5 GTGO Panel**

Indicators

Table 2-6 describes the indicators of the GPON interface cards.

Table 2-6 Indicator Description for GPON Interface Cards

Indicator	Status	Description
RUN	LED is OFF.	● The card is powered off. ● The card does not match. ● CPU is not started.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes irregularly.	The card is downloading data from the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.
ACT _i (i = 1–16)	LED is OFF.	● No ONU is configured on the GPON interface. ● The GPON interface has no optical module installed. ● The GPON interface is shutdown. ● The GPON interface is TypeB protection port.
	Green LED flashes.	The GPON interface and connected ONUs work properly.
	Red LED is ON.	The GPON interface has Loss Of Signal (LOS) alarms.

Interfaces

GPON interface cards provide 16/8 GPON interfaces.

Button

The RST button is used for resetting card.

Principle Diagram

Figure 2-6 shows the principle diagram of the GPON interface cards.

Figure 2-6 Principle Diagram of GPON Interface Cards

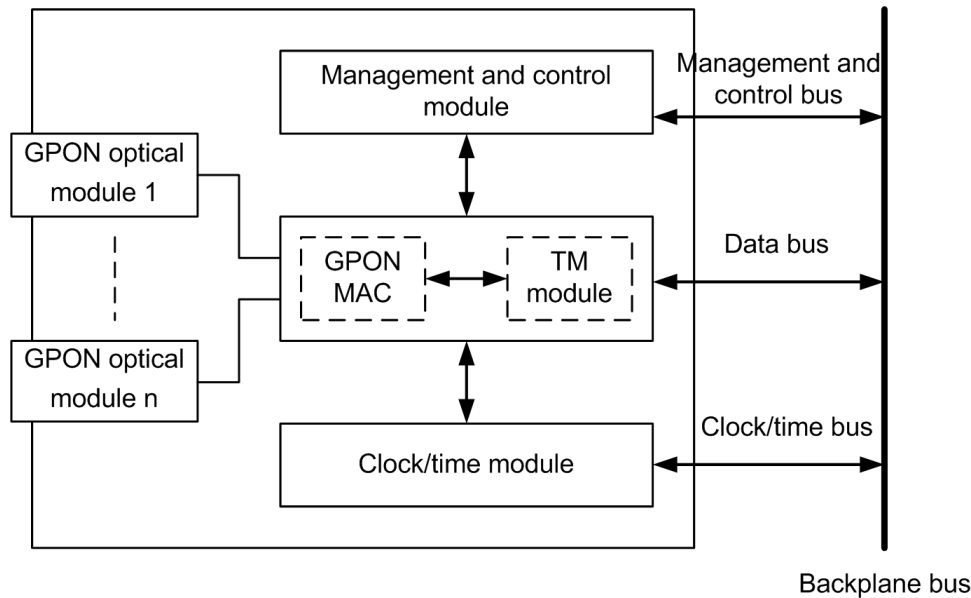


Table 2-7 describes the module functions of GPON interface cards.

Table 2-7 Module Functions of GPON Interface Cards

Module	Function
Management and control module	Manages and controls the card, implement card configuration.
GPON MAC module	Implement the GPON functions defined by ITU-T G.984.3.
TM module	Processes data in service layer, including bandwidth management and QoS process, satisfies the SLA requirement. The process function is compatible with TR156.
GPON optical module	Provides PON-C optical interface, complies with ITU-T G.984.2.
Clock/time module	Process system clock, complies with ITU-T G.8262, G.8264, and G.781.

Technical Specifications

Table 2-8 lists technical specifications of the GPON interface cards.

Table 2-8 Technical Specifications of GPON Interface Cards

Item	Specification
Power consumption	GTGH: 5 W (static), 65 W (maximum), 55 W (typical) GTGO: 5 W (static), 35 W (maximum), 30 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	GTGH: 1.000 kg GTGO: 0.925 kg

2.2.2 XG-PON1 Interface Cards

Overview

The XG-PON1 interface cards provide XG-PON access. The ZXA10 C300 supports the following XG-PON1 interface cards:

- GTXO: 8-port XG-PON1 interface card
- GTXQ: 4-port XG-PON1 interface card

Panel

Figure 2-7 and Figure 2-8 show the XG-PON1 interface card panels.

Figure 2-7 GTXO Panel

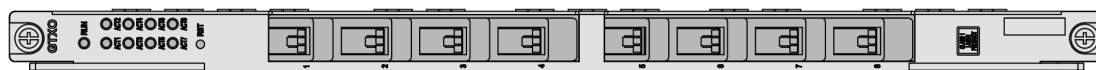
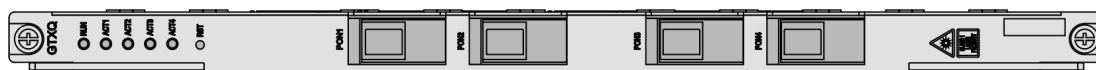


Figure 2-8 GTXQ Panel



Indicators

Table 2-9 describes the indicators of the XG-PON1 interface cards.

Table 2-9 Indicator Description for XG-PON1 Interface Cards

Indicator	Status	Description
RUN	LED is OFF.	● The card is powered off. ● The card does not match. ● CPU is not started.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes irregularly.	The card is downloading data from the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.

Indicator	Status	Description
ACT <i>i</i> (<i>i</i> = 1–8)	LED is OFF.	- No ONU is configured on the interface. - The interface has no optical module installed. - The interface is shutdown. - The interface is TypeB protection port.
	Green LED flashes.	The interface and connected ONUs work properly.
	Red LED is ON.	The interface has Loss Of Signal (LOS) alarms.

Interfaces

The XG-PON1 interface cards provide 8/4 XG-PON1 interfaces.

Button

The **RST** button is used for resetting card.

Principle Diagram

Figure 2-9 shows the principle diagram of the XG-PON1 interface cards.

Figure 2-9 Principle Diagram of XG-PON1 Interface Cards

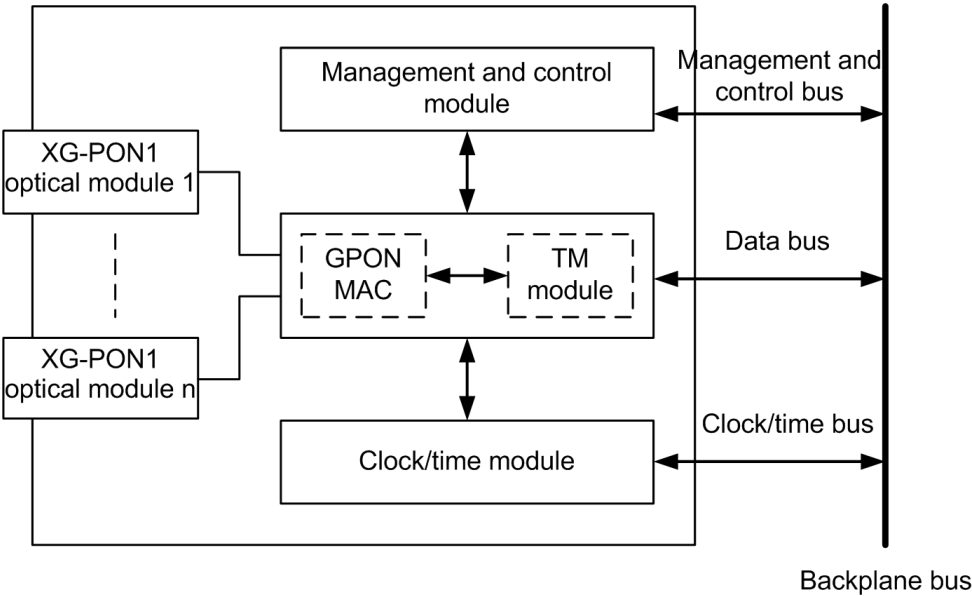


Table 2-10 describes the module functions of XG-PON1 interface cards.

Table 2-10 Module Functions of XG-PON1 Interface Cards

Module	Function
Management and control module	Manages and controls the card, implement card configuration.
XG-PON1 MAC module	Implement the XG-PON1 functions defined by ITU-T G.987.3.

Module	Function
TM module	Processes data in service layer, including bandwidth management and QoS process, satisfies the SLA requirement. The process function is compatible with TR156.
XG-PON1 optical module	Provides PON-C optical interface, complies with ITU-T G.984.2.
Clock/time module	Process system clock, complies with ITU-T G.8262, G.8264, and G.781.

Technical Specifications

Table 2-11 lists technical specifications of the XG-PON1 interface cards.

Table 2-11 Technical Specifications of XG-PON1 Interface Cards

Item	Specification
Power consumption	GT XO: 5 W (static), 100 W (maximum), 90 W (typical) GT XQ: 5 W (static), 90 W (maximum), 85 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	GT XO: 1.12 kg GT XQ: 1.19 kg

2.3 Ethernet Uplink Cards

The ZXA10 C300 uses the following Ethernet uplink cards:

- 10GE Ethernet uplink cards: XUTQ and XUVQ
- 10GE/40GE Ethernet uplink card: HUKQ
- 10GE/GE Ethernet uplink cards: HUTQ and HUVQ
- GE Ethernet uplink cards: GUFQ, GUSQ, GUTQ, and HUGQ

2.3.1 10GE Ethernet Uplink Cards

Overview

The ZXA10 C300 supports the following 10GE Ethernet uplink cards:

- XUTQ: 4-port 10GE optical interface Ethernet uplink card
- XUVQ: 4-port 10GE optical interface Ethernet uplink card, supports SynE and IEEE 1588 TC function.

Panel

Figure 2-10 and Figure 2-11 show the 10GE Ethernet uplink card panels.

Figure 2-10 XUTQ Panel

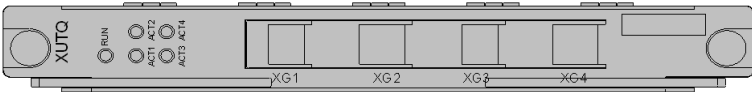
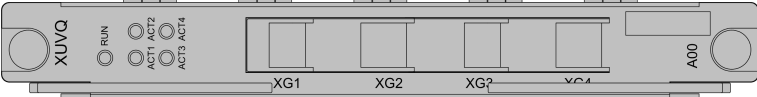


Figure 2-11 XUVQ Panel



Indicators

Table 2-12 describes the indicators for the 10GE Ethernet uplink cards.

Table 2-12 Indicator Description for 10GE Ethernet Uplink Cards

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM .
	Green LED flashes irregularly.	The card is downloading data from the switching and control card or the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.
ACT <i>i</i> (i = 1–4)	Green LED is OFF.	The 10GE optical module is not connected or optical link failure.
	Green LED is ON.	The 10GE optical interface is connected (LINK).
	Green LED flashes.	The 10GE optical interface is sending and receiving the data (ACTIVE).

Interfaces

The 10GE Ethernet uplink cards provide four 10GE Ethernet interfaces.

Principle Diagram

Figure 2-12 shows the principle diagram of the 10GE Ethernet uplink cards.

Figure 2-12 Principle Diagram of 10GE Ethernet Uplink Cards

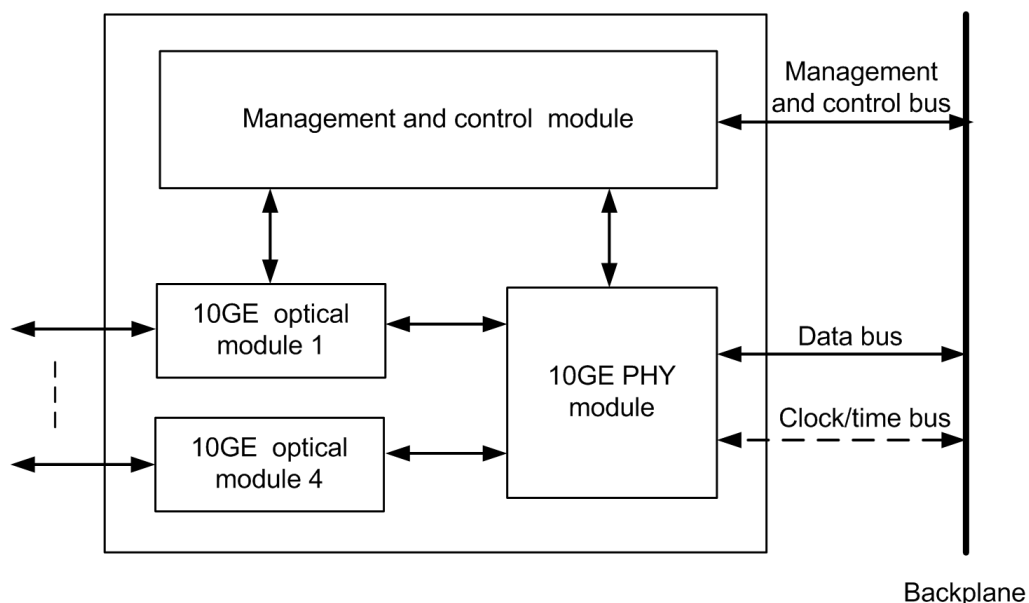


Table 2-13 describes the module functions of the 10GE Ethernet uplink cards.

Table 2-13 Module Function of 10GE Ethernet Uplink Cards

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
10GE PHY module	Implements signal connection for switching chip interface and serial Ethernet interface, coding in physical layer, clock recovery and frequency doubling. Implement time input and Ethernet frequency synchronization (only available for XUVQ).
10GE optical module	Provides 10GE optical interface.

Technical Specifications

Table 2-14 lists technical specifications of the 10GE Ethernet uplink cards.

Table 2-14 Technical Specifications of 10GE Ethernet Uplink Cards

Item	Specification
Power consumption	XUTQ: 22 W (static/maximum), 20 W (typical) XUVQ: 25 W (static/maximum), 23 W (typical)
Dimensions	197.75 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	XUTQ: 0.514 kg XUVQ: 0.445 kg

2.3.2 10GE/40GE Ethernet Uplink Card

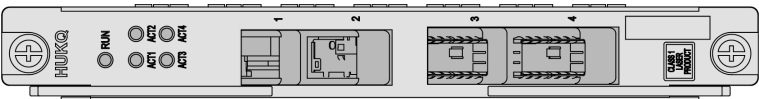
Overview

The 10GE/40GE Ethernet uplink card HUKQ provides 10GE and 40GE uplink interfaces.

Panel

Figure 2-13 shows the HUKQ panel.

Figure 2-13 HUKQ Panel



Indicators

Table 2-15 describes the indicators of the HUKQ card.

Table 2-15 HUKQ Indicator Description

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM .
	Green LED flashes irregularly.	The card is downloading data from the switching and control card or the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.
ACT <i>i</i> (<i>i</i> = 1–4)	Green LED is OFF.	The optical module is not connected or optical link failure.
	Green LED is ON.	The optical interface is connected (LINK).
	Green LED flashes.	The optical interface is sending and receiving the data (ACTIVE).

Interfaces

The HUKQ panel provides two 10GE (1 – 2) and two 40GE (3 – 4) Ethernet interfaces.

Principle Diagram

Figure 2-14 shows the principle diagram of the HUKQ card.

Figure 2-14 HUKQ Principle Diagram

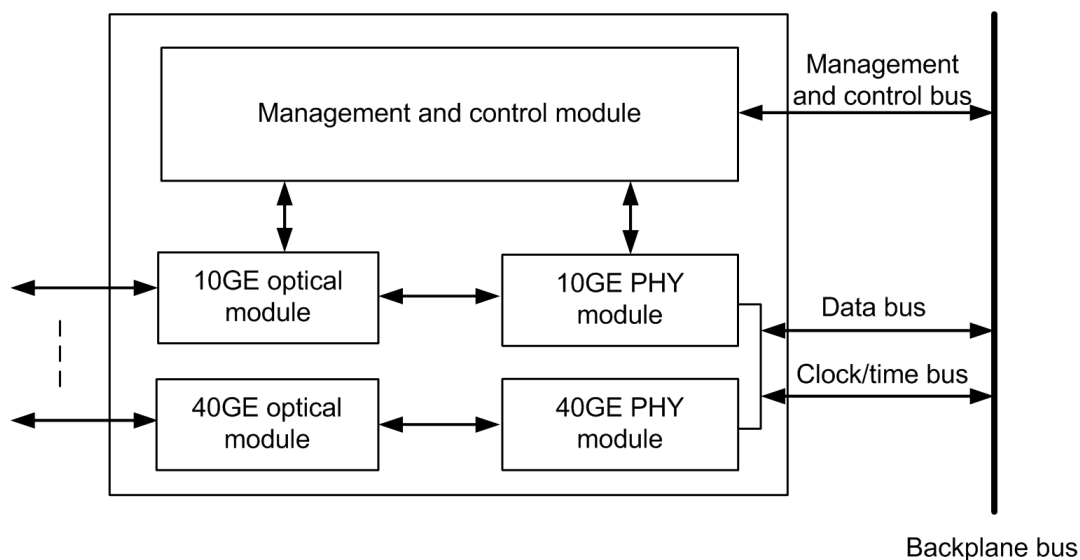


Table 2-16 describes the module functions of the HUKQ card.

Table 2-16 Module Function of HUKQ Card

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
10GE PHY module/40GE PHY module	Implements signal connection for switching chip interface and serial Ethernet interface, coding in physical layer, clock recovery and frequency doubling. Implement time input and Ethernet frequency synchronization.
10GE optical module/40GE optical module	Provides 10GE/40GE optical interface.

Technical Specifications

Table 2-17 lists technical specifications of the HUKQ card.

Table 2-17 HUKQ Technical Specifications

Item	Specification
Power consumption	22 W (static/maximum), 20 W (typical)
Dimensions	197.75 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	0.39 kg

2.3.3 10GE/GE Ethernet Uplink Cards

Overview

The ZXA10 C300 supports the following 10GE/GE Ethernet uplink cards:

- HUTQ: 2-port 10GE and 2-port GE optical interface Ethernet uplink card
- HUVQ: 2-port 10GE and 2-port GE optical interface Ethernet uplink card, supports SynE and IEEE 1588 TC function.

Panel

Figure 2-15 and Figure 2-16 show the 10GE/GE Ethernet uplink card panels.

Figure 2-15 HUTQ Panel

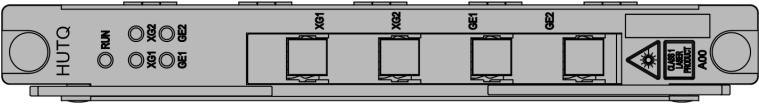
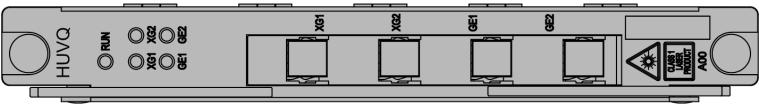


Figure 2-16 HUVQ Panel



Indicators

Table 2-18 describes the indicators of the 10GE/GE Ethernet uplink cards.

Table 2-18 Indicator Description for 10GE/GE Ethernet Uplink Cards

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes irregularly.	The card is downloading data from the switching and control card or the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.

Indicator	Status	Description
XG1/XG2/G-E1/GE2	Green LED is OFF.	The optical module is not connected or optical link failure.
	Green LED is ON.	The optical interface is connected (LINK).
	Green LED flashes.	The optical interface is sending and receiving the data (ACTIVE).

Interfaces

The 10GE/GE Ethernet uplink cards provide two 10GE optical interfaces and two GE optical interfaces.

Principle Diagram

Figure 2-17 shows the principle diagram of the 10GE/GE Ethernet uplink cards.

Figure 2-17 Principle Diagram of 10GE/GE Ethernet Uplink Cards

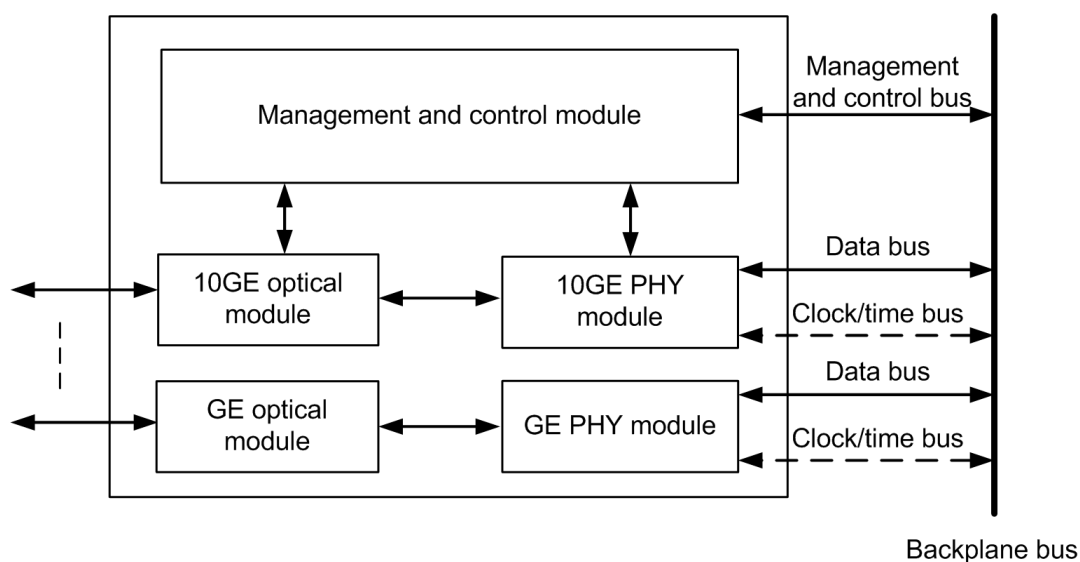


Table 2-19 describes the module functions of the 10GE/GE Ethernet uplink cards.

Table 2-19 Module Function of 10GE/GE Ethernet Uplink Cards

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
10GE PHY module/GE PHY module	Implements signal connection for switching chip interface and serial Ethernet interface, coding in physical layer, clock recovery and frequency doubling. Implement time input and Ethernet frequency synchronization (only available for HUVQ).

Module	Function
10GE optical module/GE optical module	Provides 10GE/GE optical interface.

Technical Specifications

Table 2-20 lists technical specifications of the 10GE/GE Ethernet uplink cards.

Table 2-20 Technical Specifications of 10GE/GE Ethernet Uplink Cards

Item	Specification
Power consumption	15 W (static/maximum), 13 W (typical)
Dimensions	197.75 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	0.42 kg

2.3.4 GE Ethernet Uplink Cards

Overview

The ZXA10 C300 supports the following GE Ethernet uplink cards:

- **GUFQ:** 4-port optical interface Ethernet uplink card
Four optical interfaces can be GE optical interface (a GFSQ sub-card needs to be configured), or FE optical interface (an FFSQ sub-card needs to be configured).
- **GUSQ:** 2-port optical and 2-port electrical interface Ethernet uplink card
Two optical interfaces can be GE optical interface (with a GFSD sub-card), or FE optical interface (with an FFSD sub-card).
Two electrical interfaces are 10 M/100 M/1000 M self-adaptive electrical interfaces (with a GTSD sub-card).
- **GUTQ:** 4-port electrical interface Ethernet uplink card
Four electrical interfaces are 10 M/100 M/1000 M self-adaptive electrical interfaces.
- **HUGQ:** 2-port GE and 2-port FE optical interface Ethernet uplink card

Panel

Figure 2-18, Figure 2-19, Figure 2-20, and Figure 2-21 show the GE Ethernet uplink card panels.

Figure 2-18 GUFQ Panel

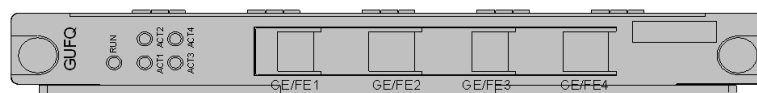


Figure 2-19 GUSQ Panel

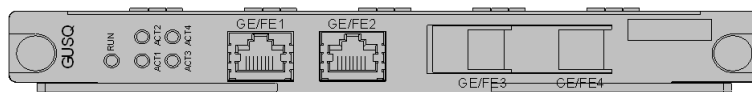


Figure 2-20 GUTQ Panel

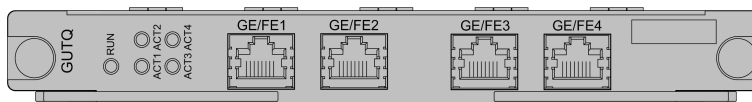
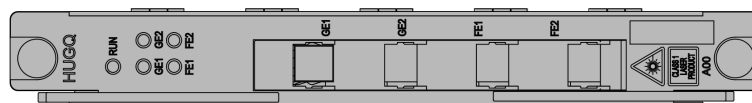


Figure 2-21 HUGQ Panel



Indicators

Table 2-21 describes the indicators of the GE Ethernet uplink cards.

Table 2-21 Indicator Description for GE Ethernet Uplink Card

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes irregularly.	The card is downloading data from the switching and control card or the network management server.
	Green LED flashes slowly (0.5 Hz).	The card runs properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card is plugged in the wrong slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. Auto-update is not configured on the card.
ACT _i (i = 1–4)	Green LED is OFF.	The interface is not connected.
	Green LED is ON.	The interface is connected (LINK).
	Green LED flashes.	The interface is sending and receiving the data (ACTIVE).

Interfaces

The GE Ethernet uplink cards provide four GE/FE interfaces.

Principle Diagram

Figure 2-22 shows the principle diagram of the GE Ethernet uplink cards.

Figure 2-22 Principle Diagram of GE Ethernet Uplink Card

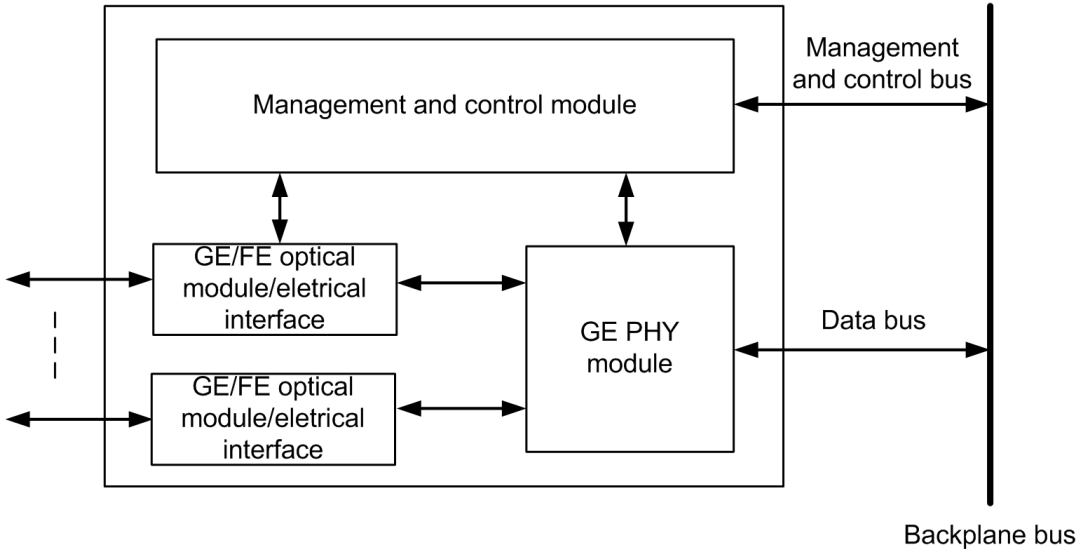


Table 2-22 describes the module functions of the GE Ethernet uplink cards.

Table 2-22 Module Function of GE Ethernet Uplink Cards

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
GE PHY module	Implements GE serial Ethernet interface conversion, coding in physical layer, clock recovery and frequency doubling.
GE/FE optical module/electrical interface	Provides GE/FE optical/electrical interface.

Technical Specifications

Table 2-23 lists technical specifications of the GE Ethernet uplink cards.

Table 2-23 Technical Specifications of GE Ethernet Uplink Card

Item	Specification
Power consumption	5 W (static/maximum), 4 W (typical)
Dimensions	197.75 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	GUFQ: 0.506 kg GUSQ: 0.466 kg GUTQ: 0.480 kg HUGQ: 0.400 kg

2.4 Ethernet Interface Cards

Overview

The ZX A10 C300 supports the following Ethernet interface cards:

- GDFO: 8-port GE Ethernet interface card
- HDTT: 8-port GE and 2-port 10GE Ethernet interface card

Ethernet interface cards support the link aggregation within a card or among cards based on [LACP](#).

Panel

[Figure 2-23](#) and [Figure 2-24](#) show the Ethernet interface card panels.

Figure 2-23 GDFO Panel

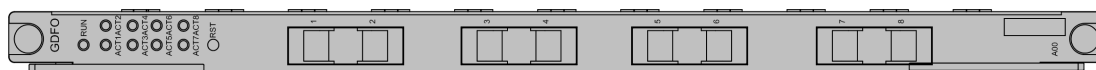
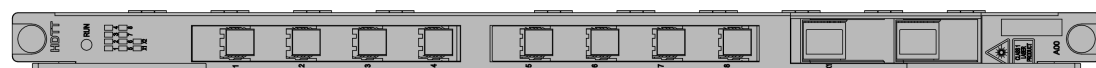


Figure 2-24 HDTT Panel



Indicators

[Table 2-24](#) describes the indicators of the Ethernet interface cards.

Table 2-24 GDFO Indicator Description

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off or CPU is not started.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM .
	Green LED flashes quickly.	The card is downloading data from the switching and control card or NM.
	Green LED flashes slowly (0.5 Hz).	The card is working properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	This card should not be inserted in this slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card does not match that of the switching and control card. The service is normal.

Indicator	Status	Description
ACT <i>i</i> (<i>i</i> = 1–8) Y1–8/X1–X2	Green LED is OFF.	The optical interface is not connected.
	Green LED is ON.	The optical interface is connected (LINK).
	Green LED flashes.	The optical interface is sending and receiving the data (ACTIVE).

Interfaces

The Ethernet interface cards provides eight GE optical interfaces (1 – 8) and two 10GE optical interfaces (X1 – X2, only valid for HDTT).

Button

The **RST** button is used for resetting card.

Principle Diagram

Figure 2-25 shows the principle diagram of the Ethernet interface cards.

Figure 2-25 Principle Diagram of Ethernet Interface cards

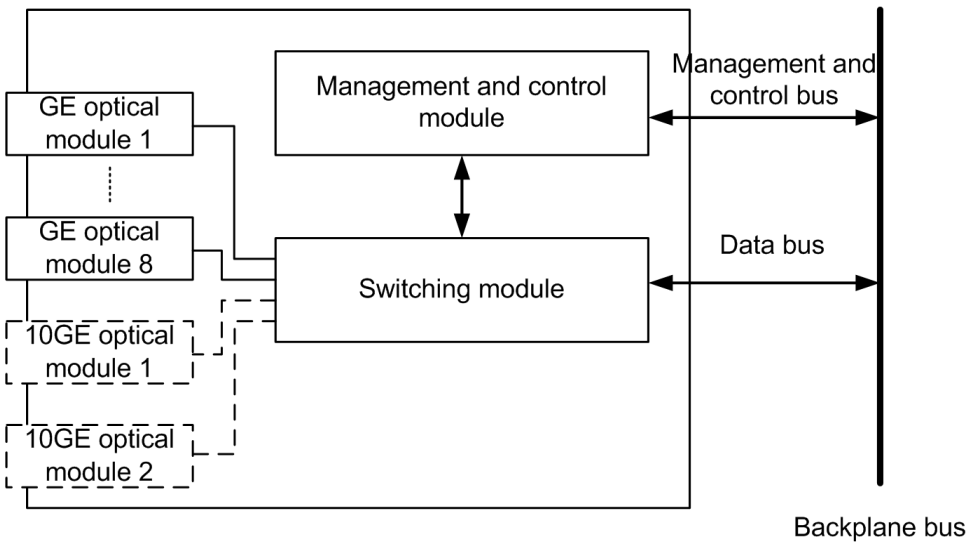


Table 2-25 describes the module functions of the Ethernet interface cards.

Table 2-25 GDFO Module Function of Ethernet Interface Cards

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
Switching module	Implements data conversion for optical module and backplane bus, implement VLAN and Ethernet functions.
GE/10GE optical module	Provides GE/10GE optical interface.

Technical Specifications

Table 2-26 lists technical specifications of the Ethernet interface cards.

Table 2-26 Technical Specifications of Ethernet Interface Cards

Item	Specification
Power consumption	GDFO: 5 W (static), 30 W (maximum), 27 W (typical) HDTT: 5 W (static), 35 W (maximum), 32 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	GDFO: 0.755 kg HDTT: 0.77 kg

2.5 P2P Interface Card

Overview

The FTGK card is a 48-port **P2P** interface card. It uses **WDM** technology and single optical fiber for transmitting and receiving.

Panel

Figure 2-26 shows the FTGK panel.

Figure 2-26 FTGK Panel



Indicator

Table 2-27 describes the indicators of the FTGK card.

Table 2-27 FTGK Indicator Description

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off, or CPU is not started.
	Green LED is ON.	- The card passes the self test, but fails to obtain data from the switching and control card or NMS. - The card is running the booting program, or is downloading a software version from the switching and control card. - The card cannot operate because there is no legal software version.
	Green LED flashes quickly.	The card is downloading data from the switching and control card or the NMS.

Indicator	Status	Description
	Green LED flashes slowly (0.5 Hz).	The card is operating properly.
	Red LED is ON.	- A serious fault occurs to the hardware of the card (such as, the card fails to pass the self test, or the software version is incompatible with the hardware version). - The NMS is sending a restarting command.
	Red LED flashes slowly (0.5 Hz).	This card should not be inserted in this slot (different from the configuration of the NMS).
	Yellow LED flashes slowly (0.5 Hz).	The software version of the card is incompatible with that of the switching and control card (auto-update function is not configured on the card). The services are proper.
LED _i (i = 1–48)	Green LED is OFF.	No. i GE/FE optical port is not connected.
	Green LED is ON.	No. i GE/FE optical port is connected (LINK).
	Green LED flashes.	No. i GE/FE optical port is sending and receiving the data (ACTIVE).

Interface

The FTGK card provides 48 FE/GE optical interfaces (24 ports, each port provides 2 interfaces).

Principle Diagram

Figure 2-27 shows the principle diagram of the FTGK card.

Figure 2-27 FTGK Principle Diagram

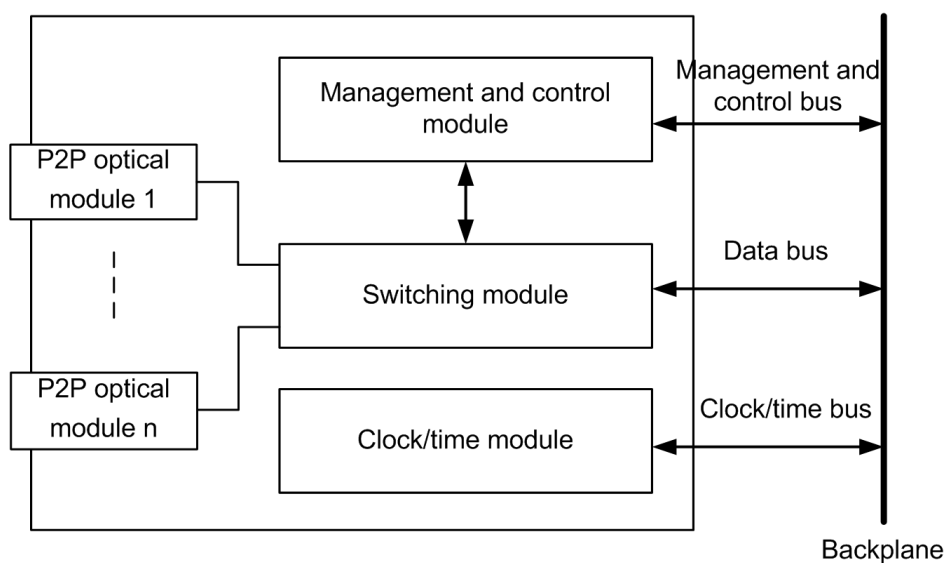


Table 2-28 describes the module functions of the FTGK card.

Table 2-28 FTGK Module Function

Module	Function
Management and control module	Implements card configuration, traffic management and optical module management, and status detection of optical module.
Switching module	Implements conversion of optical module and backplane bus, implements VLAN and Ethernet functions.
Clock module	Supports all optical interface to use system clock for clock synchronization. When the card is used for uplink, the synchronized clock from the optical port is sent to the switching and control card works as the system clock.
P2P optical module	Provides GE/FE optical interface.

Technical Specification

Table 2-29 lists technical specifications of the FTGK card.

Table 2-29 FTGK Technical Specification

Item	Specification
Power consumption	5 W (static), 100 W (maximum), 85 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	1.51 kg

2.6 TDM Interface Cards

The ZX A10 C300 uses the following TDM interface cards.

- STM-N CES interface card: CTLA
- E1 CES interface cards: CTUB

2.6.1 STM-N CES Interface Card

Overview

The CTLA card is the STM-N CES interface card, it supports TDM service over Ethernet.

Panel

Figure 2-28 shows the CTLA panel.

Figure 2-28 CTLA Panel



Indicators

Table 2-30 lists the indicator descriptions for the CTLA card.

Table 2-30 CTLA Indicator Description

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off or CPU is not started.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes quickly.	The card is downloading data from the switching and control card or NM.
	Green LED flashes (0.5 Hz).	The card is working properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card should not be inserted in this slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of line card does not match that of the switching and control card.
MOD	Green LED is OFF.	The card is in STM-1 interface (155 M) status.
	Green LED is ON.	The card is in STM-4 interface (622 M) status.
TX/RX	Two green LEDs are ON.	Optical module is ON. Data transmission and receiving is normal.
	Two green LEDs are OFF.	Optical module is OFF without being configured.
	RX Red LED is ON.	LOS or LOF alarm, or multiplex section alarm.
	TX Red LED is ON.	The remote end is invalid.

Interfaces

The CTLA card provides two STM-1 interfaces or one STM-4 interface.

Button

The **RST** button is used for resetting card.

Principle Diagram

Figure 2-29 shows the principle diagram of the CTLA card.

Figure 2-29 CTLA Principle Diagram

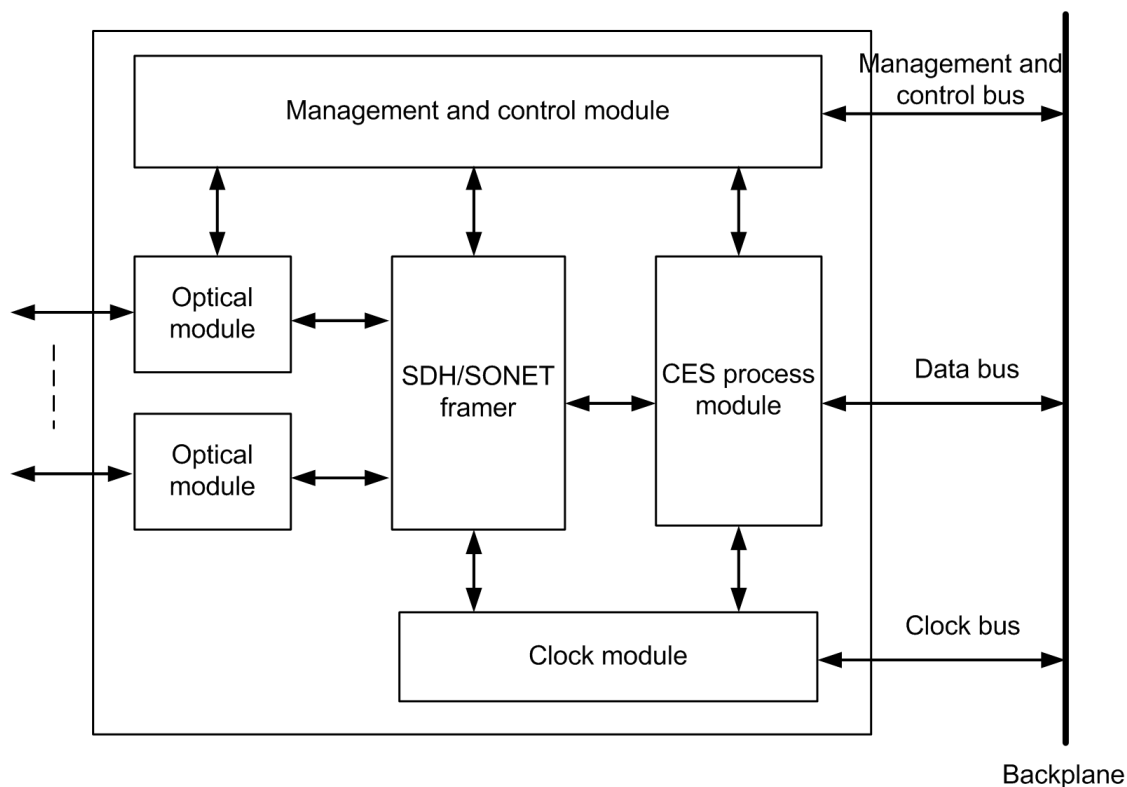


Table 2-31 describes the module functions of the CTLA card.

Table 2-31 CTLA Module Functions

Module	Function
Management and control module	Implements configuration, control, and management for each module in the card.
CES processing module	Encapsulates TDM service data to Ethernet packets and decapsulates Ethernet packets to TDM service data, extracts and restores line clock.
SDH/SONET framer	Processes overheads and alarms of the SDH/SONET optical line.
Optical module	Provides STM-1/STM-4 optical interface.

Technical Specifications

Table 2-32 lists technical specifications of the CTLA card.

Table 2-32 CTLA Technical Specifications

Item	Specification
Power consumption	35 W (static/maximum), 34 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	0.752 kg

2.6.2 E1 CES Interface Cards

Overview

The E1 CES interface card CTUB supports TDM service over Ethernet.

Panel

Figure 2-30 shows the CTUB panel.

Figure 2-30 CTUB Panel



Indicators

Table 2-33 describes the indicators of the CTUB card.

Table 2-33 Indicator Description for CTUB Card

Indicator	Status	Description
RUN	LED is OFF.	The card is powered off or CPU is not started.
	Green LED is ON.	The card self-checking completes but fails to receive data from the switching and control card or NM.
	Green LED flashes quickly.	The card is downloading data from the switching and control card or NM.
	Green LED flashes slowly.	The card is working properly.
	Red LED is ON.	Card hardware failure
	Red LED flashes slowly (0.5 Hz).	The card should not be inserted in this slot.
	Yellow LED flashes slowly (0.5 Hz).	The software version of line card does not match that of the switching and control card. The service is normal.
ALM	LED is OFF.	Tributary is not configured on NM.
	Green LED is ON.	Tributary is proper.
	Red LED is ON.	High order alarm occurs on service line from TAA to 2M card.
	Yellow LED is ON.	The configured tributary E1 line alarm occurs.

Interfaces

The CTUB card provides 32-channel unbalanced E1. Each E1 channel has four pins. The upper two pins are used for receiving and the lower two pins are used for sending.

Button

The **RST** button is used for resetting card.

Principle Diagram

Figure 2-31 shows the principle diagram of the CTUB card.

Figure 2-31 Principle Diagram of CTUB Card

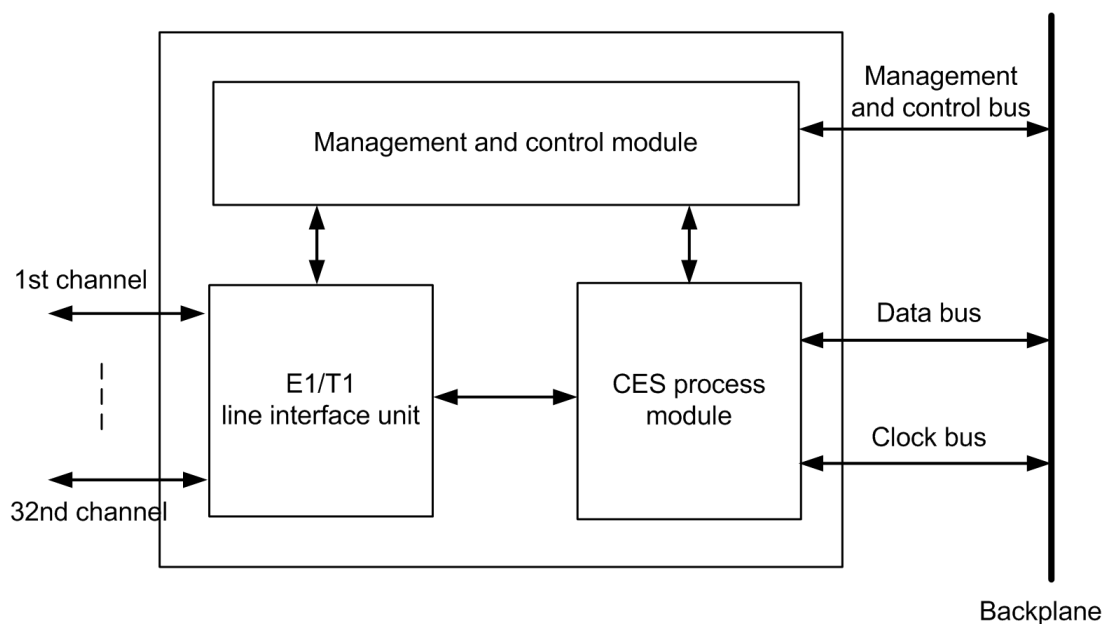


Table 2-34 describes the module functions of the CTUB card.

Table 2-34 Module Functions of CTUB Card

Module	Function
Management and control module	Implements configuration, control, and management for each module in the card.
CES processing module	Downstream: extracts and recovers clock from LIU, composes CES packets using LIU data and transmits CES packets using LIU clock. Upstream: extracts and restores clock from CES packets, retrieves data from received CES packets and transmits data to LIU using the clock in CES packets.
E1/T1 line interface unit	Implements E1/T1 line CODEC, extracts and restores clock from the E1/T1 lines.

Technical Specifications

Table 2-35 lists technical specifications of the CTUB card.

Table 2-35 Technical Specifications of CTUB Card

Item	Specification
Power consumption	16 W (static/maximum), 14 W (typical)
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	0.68 kg

2.7 Common Interface Card

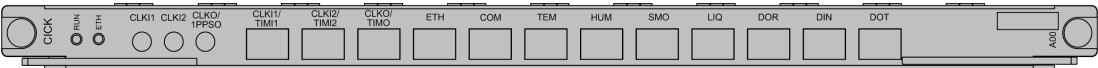
Overview

The common interface card CICK provides monitoring interfaces regarding environmental parameters, such as temperature, humidity, liquid, entrance control, and smog, as well as BITS clock and 1PPS+TOD time input and output interfaces. CICK also provides input/output ports, such as network port, serial port, and trunk nodes. These ports are used to monitor external devices.

Panel

Figure 2-32 shows the CICK panel.

Figure 2-32 CICK Panel



Indicators

Table 2-36 describes the indicators of the CICK card.

Table 2-36 CICK Indicator Description

Indicator	Status	Description
RUN	Green LED flashes.	The card is working properly.
	Red LED is ON.	Card hardware failure
ETH	Green LED is ON.	The interface connection is proper, but no data is sending or receiving.
	Green LED flashes.	The interface is sending or receiving data.

Interfaces

Table 2-37 describes the interfaces of the CICK card.

Table 2-37 CICK Interface Description

Interface	Description
CLKI1	The first BITS clock input interface, 75Ω
CLKI2	The second BITS clock input interface, 75Ω
CLKO/1PPSO	BITS/1PPS+TOD clock output interface, 75Ω
CLKI1/TIMI1	The first BITS/1PPS+TOD clock input interface, 120Ω
CLKI2/TIMI2	The second BITS/1PPS+TOD clock input interface, 120Ω
CLKO/TIMO	BITS/1PPS+TOD clock output interface, 120Ω
ETH	Out-of-band maintenance network interface
COM	Public serial port
TEM	Temperature sensor interface
HUM	Humid sensor interface
SMO	Smog sensor interface
LIQ	Liquid sensor interface
DOR	Entrance control sensor interface
DIN	Digit input interface, four channels
DOT	Digit input interface, four channels

Principle Diagram

Figure 2-33 shows the principle diagram of the CICK card.

Figure 2-33 CICK Principle Diagram

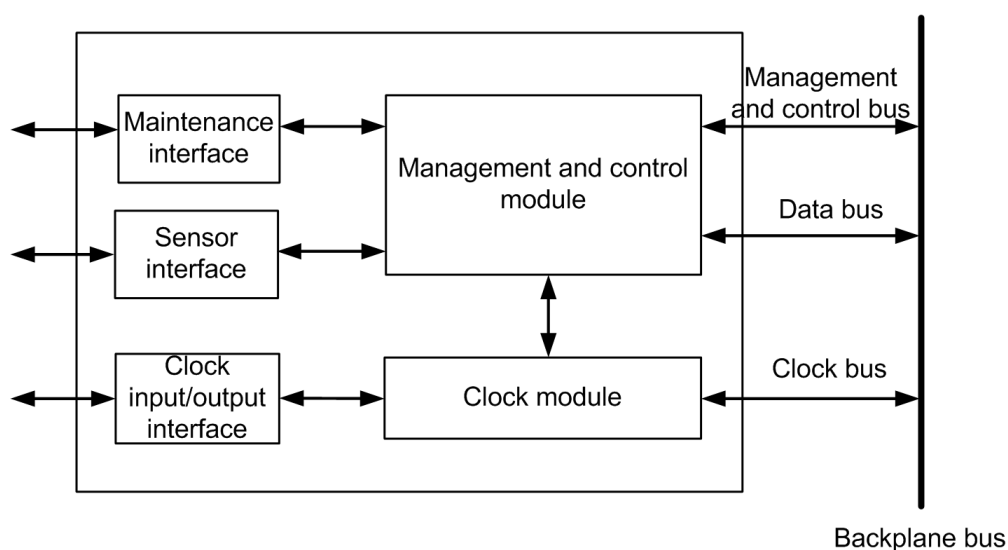


Table 2-38 describes the module functions of the CICK card.

Table 2-38 CICK Module Function

Module	Function
Management and control module	Implements configuration, control, and management for each module in the card.
Interface module	Provides maintenance interfaces, sensor interfaces and clock input/output interfaces.
Clock module	Processes input and output clock on 75Ω/120Ω interfaces.

Technical Specifications

Table 2-39 lists technical specifications of the CICK card.

Table 2-39 CICK Technical Specifications

Item	Specification
Power consumption	5 W
Dimensions	395.5 mm × 22.5 mm × 225 mm (Height × Width × Depth)
Weight	0.75 kg

2.8 Power Card

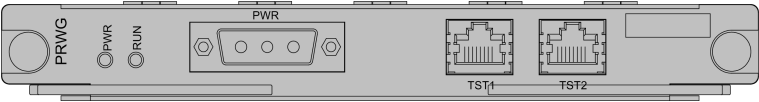
Overview

The power card PRWG card uses –48 V DC power and provides power supply to each card. The system supports two power cards. When the input voltage of two power supplies is equal, the two PRWG cards work in load-balance mode.

Panel

Figure 2-34 shows the PRWG card.

Figure 2-34 PRWG Panel



Indicators

Table 2-40 describes the indicators of the PRWG card.

Table 2-40 PRWG Indicator Description

Indicator	Status	Description
PWR	Green LED is ON.	Power is ON.
RUN	Green LED is ON.	The card is running properly.

Interfaces

Table 2-41 describes the interfaces of the PRWG card.

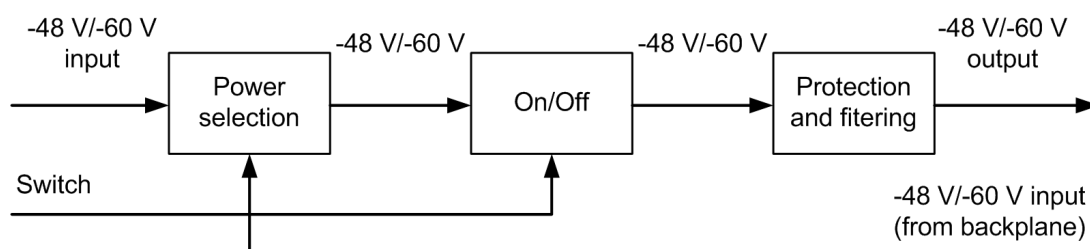
Table 2-41 PRWG Interface Description

Interface	Description
PWR	Power interface
TST1/TST2	Reserved test interface

Principle Diagram

Figure 2-35 shows the principle diagram of the PRWG card.

Figure 2-35 PRWG Principle Diagram



Technical Specifications

Table 2-42 lists technical specifications of the PRWG card.

Table 2-42 PRWG Technical Specifications

Item	Specification
Power consumption	5 W
Dimensions	197.75 mm × 25 mm × 225 mm (Height × Width × Depth)
Weight	0.36 kg

2.9 Backplanes

The ZX A10 C300 supports the following backplanes:

- 21-inch backplane: MWEA and MWEA/R
- 19-inch backplane: MWIA and MWIA/R

2.9.1 21-Inch Backplane

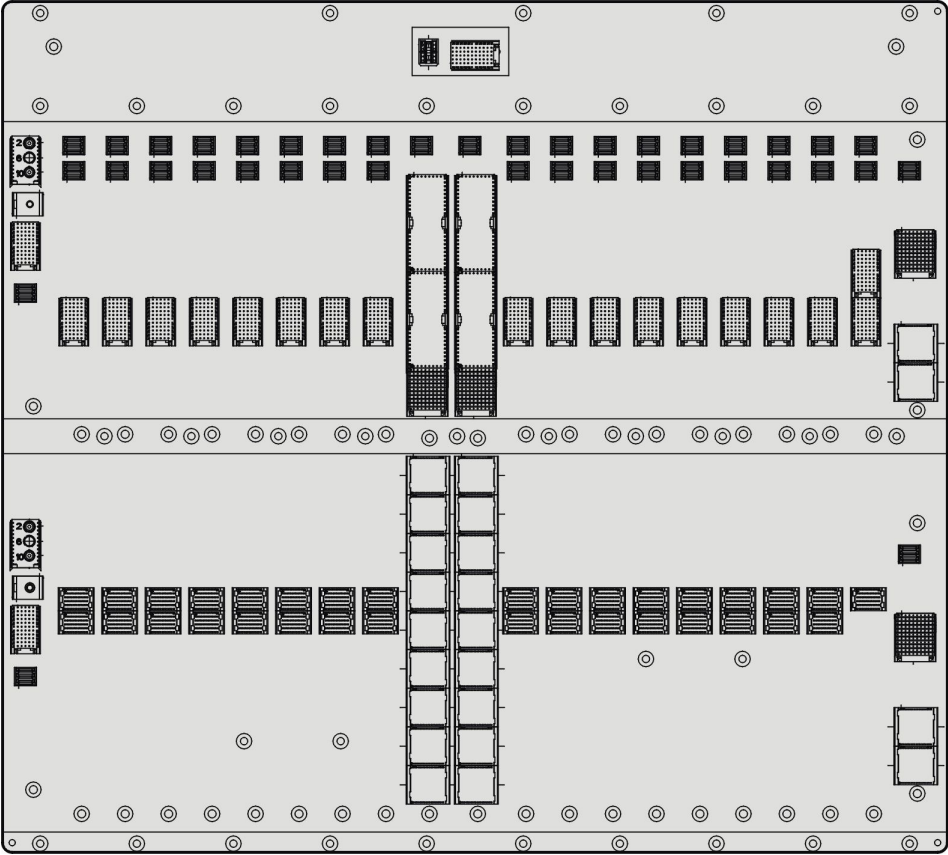
Overview

The 21-inch MWEA or MWEA/R backplane is used with the ETSI 21-inch cabinet.

Panel

Figure 2-36 shows the 21-inch backplane.

Figure 2-36 21-Inch Backplane



Interfaces

The backplane provides the following interfaces:

- Interfaces for switching and control cards, interface cards, and uplink cards
- Power card interface: -48 V, 48 V GND, 3.3 V, and GND
- Fan socket: -48 V, -48 V GND, and fan detection/control

Technical Specifications

Table 2-43 lists technical specifications of the 21-inch backplane.

Table 2-43 21-Inch Backplane Technical Specifications

Item	Specification
Power consumption	-
Dimensions	440 mm × 481 mm (Height × Width)
Weight	2.57 kg

2.9.2 19-Inch Backplane

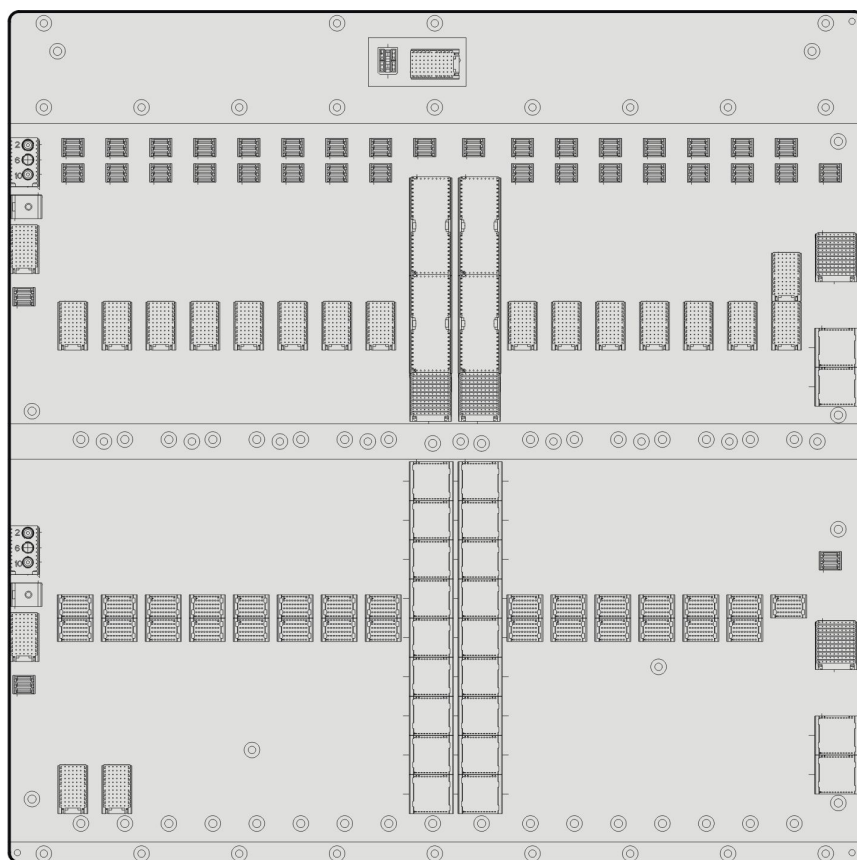
Overview

The 19-inch MWIA or MWIA/R backplane is used with the IEC 19-inch cabinet.

Panel

Figure 2-37 shows the 19-inch backplane.

Figure 2-37 19-Inch Backplane



Interfaces

The backplane provides the following interfaces:

- Interfaces for switching and control cards, interface cards, and uplink cards
- Power card interface: -48 V, 48 V GND, 3.3 V, and GND
- Fan socket: -48 V, -48 V GND, and fan detection/control

Technical Specifications

Table 2-44 lists technical specifications of the 19-inch backplane.

Table 2-44 19-Inch Backplane Technical Specifications

Item	Specification
Power consumption	-
Dimensions	440 mm × 440 mm (Height × Width)
Weight	2.35 kg

2.10 Fan Boxes

Overview

The ZXA10 C300 uses a fan box for heat dissipation. The fan box consists of a fan control card and several fans. Two types of fan control cards are available: FCWB and FCWD.

The fan control card supports the following functions:

- Controls the rotating speed of fans to cool the system in accordance with the requirements for the working temperature.
- Indicates the operating status of fans.
- Controls the rotating speed of fans in accordance with the commands received from a [NMS](#).
- Reports the status of each fan to the NMS.

[Table 2-45](#) lists the differences between the FCWB and the FCWD cards.

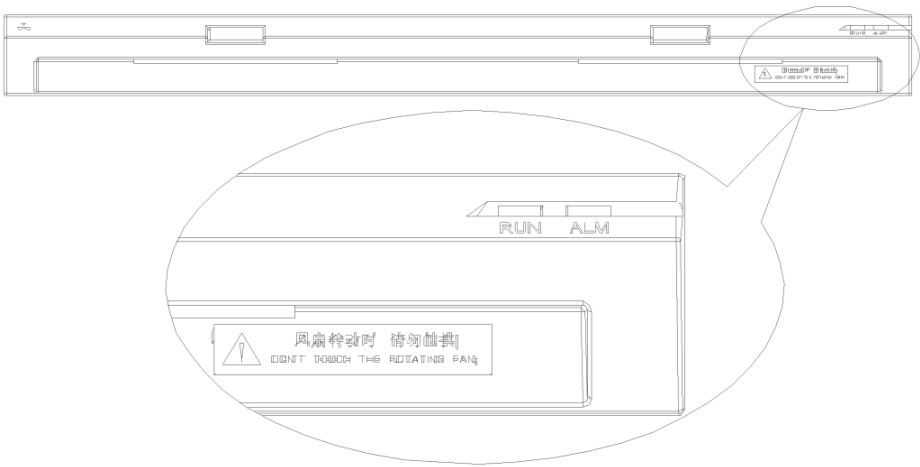
Table 2-45 FCWB and FCWD Cards

Item	FCWB	FCWD
Number of fans	Fan-19: 3 Fan-21: 4	Fan-19: 10 Fan-21: 10
Fan dimensions	127 mm × 127 mm × 38 mm	80 mm × 80 mm × 38 mm
Target rotating speed	4800 RPM	7500 RPM

Panel

[Figure 2-38](#) displays the panel of the fan box.

Figure 2-38 Panel of the Fan Box



Indicators

Table 2-46 describes the indicators of the fan boxes.

Table 2-46 Fan Boxes Indicator Description

Indicator	Status	Description
RUN	Green LED is ON.	Fans are working properly.
ALM	Red LED flashes.	One or more fans are faulty.

Technical Specifications

Table 2-47 lists technical specifications of fan boxes.

Table 2-47 Fan Boxes Technical Specifications

Item	Specification
Power consumption	FAN-19 (FCWB): 44 W FAN-21 (FCWB): 55 W FAN-19/21 (FCWD): 55 W (typical, 60% speed), 130 W (maximum)
Dimensions (Height × Width × Depth)	FAN-19 (FCWB): 43.6 mm × 437 mm × 258.1 mm FAN-19 (FCWD): 43.6 mm × 437 mm × 258.7 mm FAN-21 (FCWB): 43.6 mm × 490.5 mm × 258.1 mm FAN-21 (FCWD): 43.6 mm × 490.5 mm × 258.7 mm
Weight	FAN-19 (FCWB): 2.5 kg FAN-19 (FCWB): 3.0 kg FAN-21 (FCWD): 2.7 kg FAN-21 (FCWD): 3.2 kg

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Glossary

1PPS

- 1 Pulse Per Second

BITS

- Building Integrated Timing Supply

CES

- Circuit Emulation Service

CES

- Channel Element Subsystem

CODEC

- Code and Decode

CPU

- Central Processing Unit

DC

- Direct Current

FE

- Fast Ethernet

GE

- Gigabit Ethernet

GPON

- Gigabit Passive Optical Network

LACP

- Link Aggregation Control Protocol

LIU

- Line Interface Unit

LOF

- Loss of Frame

LOS

- Loss Of Signal

MAC

- Media Access Control

NM

- Network Management

NMS

- Network Management System

P2P

- Point to Point

PON

- Passive Optical Network

RS-232

- Recommended Standard 232

SLA

- Service Level Agreement

STM

- Synchronous Transport Module

STM-N

- Synchronous Transport Module, level N(N=1, 4, 16, 64)

TC

- Transparent Clock

TDM

- Time Division Multiplexing

TM

- TrafficManagement

TOD

- Time of Day

WDM

- Wavelength Division Multiplexing